Bhavesh Ghanchi

Software Engineer | Distributed Systems & Backend

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EXPERIENCE

Easley-Dunn Productions **San Francisco, CA**

Software Engineer Intern | Co-Lead, Multiplayer Core Gameplay *Jul 2026 - Present*

- Architected a reusable **C#** vehicle-simulation and **state-machine framework** that unified Robot, Car, and Plane handling while centralizing physics tuning, transformations, drifting, combat, and recovery across 6 tracks.
- Profiled the multiplayer networking layer to root-cause a client-server tick mismatch, then rebuilt fault-tolerant state sync for 6-player races, delivering **67% fewer transform updates** at 10 Hz against a 30 Hz tick with connectivity recovery.

Streets Are For Everyone (SAFE) **Los Angeles, CA**

Software Developer Intern *Jun 2025 - Aug 2025*

- Architected a geospatial intelligence platform using **React, Node.js, and Python**, designing an ETL and graph-construction pipeline that converted 497 GIS features into 1,248 routable segments, increasing routing granularity by **151%** with **100% coverage**.
- Launched an **LLM**-assisted route planner for a 60-cyclist beta, extracting safety preferences from natural-language requests, grounding recommendations in GIS data, and iterating on rider feedback before wider rollout.

Colgate-Palmolive **Mumbai, India**

Full-Stack Intern *Jan 2024 - Jun 2024*

- Built a scalable **React** and **TypeScript** design system with 25+ reusable components, still in production on the Hill's Pet Hub, standardizing component architecture and frontend delivery across **7 international markets**.
- **Automated** multi-environment **CI/CD** releases, improving reliability for a launch drawing **100,000+ visits and 20,000 unique logins** on day one, then cut peak API **latency 25%** by profiling slow queries and adding Redis caching.

K. J. Somaiya College of Engineering **Mumbai, India**

Research Intern *Mar 2023 - May 2023*

- Built a Python clinical **NLP pipeline** processing **5,000+ healthcare documents**, fine-tuning **BERT with SciSpacy** to extract medical entities and classify content across 12 categories, achieving an **89% F1 score**.

PROJECTS

NimbusKV: Replicated Key-Value Store | github.com/bhaveshGhanchi/NimbusKV **Jul 2026**

Go (standard library only), HTTP, Docker Compose, 25 unit + integration tests

- Built a 3-node distributed key-value store in **Go** using only the standard library, with Raft-style leader election and **quorum** writes, sustaining **~100 ms median failover** on an 80–160 ms election timeout.
- Designed the storage architecture around atomic snapshots and tombstone deletes, **benchmarking replication** at **2.4x single-node write latency** (70 → 167 µs p50), **1.3 ms prefix scans** over 10k keys, and **5.6 ms** snapshots.

OTAS: Observability Tool for Agentic AI Systems | github.com/bhaveshGhanchi/otas **Apr 2026**

Python, Django, PostgreSQL, Redis, Celery, OpenTelemetry, MCP, Docker

- Co-engineered an **observability** and **monitoring** platform for agentic AI systems, capturing prompts, tool calls, latency, and outcomes while reconstructing execution lineage across multi-agent runs, handoffs, retries, and failures.
- Built **OpenTelemetry** tracing and a Pydantic-validated **MCP** ingestion server with CLI hooks for Claude Code, Cursor, and Codex; load-tested 10 concurrent agents across **800 sessions** at **149 ms p95 latency**.

LEAP: Loss-Aware End-to-End Acknowledged Protocol | github.com/bhaveshGhanchi/leap **May 2026**

Java, UDP Sockets, Maven

- Implemented a TCP-style reliable transport protocol from scratch over **UDP** with sliding windows, cumulative ACKs, fast retransmission, adaptive RTO, Tahoe **congestion control**, and CRC32/SHA-256 integrity checks.
- Built an automated benchmarking harness with 3 packet-loss simulation modes, profiling retransmission behavior to locate throughput bottlenecks while sustaining **46.2 MB/s** at 0% loss and **88.8% efficiency** at 10% loss.

EDUCATION

University of Southern California, Los Angeles, CA | MS Computer Science | GPA: 3.83/4.00 May 2026

K. J. Somaiya College of Engineering, Mumbai, India | BTech Computer Engineering | CGPA: 9.64/10 Jun 2024

SKILLS

Languages: Python, Java, Go, C#, TypeScript, JavaScript, SQL

AI & ML: Machine Learning, LLMs, RAG, Transformers, BERT, PyTorch, Embeddings, FAISS, QLoRA, Hugging Face, Agentic Workflows, MCP

Frameworks: React, Django, FastAPI, Node.js, Express, Celery, REST APIs, Microservices

Data & Infrastructure: PostgreSQL, Redis, Firebase, Docker, CI/CD, Git, Linux, OpenTelemetry

Systems: Distributed Systems, Concurrency, Replication & Consensus, Performance Optimization, Networking (TCP/UDP/HTTP), BGP

PUBLICATIONS

M. Garg, L. Wang, **B. Ghanchi**, et al. "Biomedical Literature Q&A System Using RAG." arXiv:2509.05505, Sep 2025.

- Co-developed a biomedical **RAG** question-answering system using MiniLM embeddings, top-5 FAISS retrieval, and QLoRA-tuned Mistral-7B on MedQuAD, achieving **0.88 - 0.90 BERTScore F1**.